

The Obstruction Calculus, Consolidated

The Λ -Corpus Definitional Kernel, Formalized, with the Hodge Instrument as Flagship Instantiation

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0. Purpose and reading instructions

The repository contains, distributed across four principal documents — *Semantic Obstruction and Irreducible Residue* v1.2 (SOIR), its Appendixes B/C/D, *Residue-Aware Computation as Groupoid Traversal*, and *Bounded Formal Systems, Traversal Claims, and the Structural Malformation of Universality Arguments* (BFS) — a definitional calculus for obstruction that is more rigorous than its lab-notes framing advertises. This consolidation does four things:

1. **Extracts the kernel:** the minimal set of definitions and theorems that carry all the weight, stated once, with the corpus's own internal cross-references resolved.
2. **Grades and corrects:** separates the rigorous kernel from Tier-III material, and records the corrections a referee would demand.
3. **Proves the flagship instantiation:** the Hodge traversal instrument $\mathfrak{B}/\mathfrak{Q}$ is exhibited as a full worked instance of the calculus — and it turns out to be a *stronger* instance than any example currently in the corpus, yielding three new theorems: uniform quantitative separation with explicit constant, the vanishing identity for apparent obstruction (which is the exact formal content of "iteration is semantic"), and a historical strict-monotonicity witness of Millennium-problem grade.
4. **Closes the loop:** shows that SOIR and BFS, read jointly, *derive* the discipline that the Hodge Compiled Edition enforced — the corpus's own kernel proves that absolute irreducibility claims are not well-formed traversal claims. The framework and the traversal are one body of work, and this document is the joint.

Tier labels as in the compiled edition. Everything in Sections 1–2 is the corpus's content, verified; Section 3's corrections are editorial findings; Section 4's theorems are new.

1. The kernel, extracted

1.1 Primitive data (SOIR §2)

A state space X ; a target T ; for each $x \in X$ a nonempty class $R(x)$ of representations; a representation groupoid G whose objects are representations and whose invertible morphisms are admissible re-descriptions, with $R(x)$ an orbit; and an obstruction functional

$$O : \{(r, x) : x \in X, r \in R(x)\} \rightarrow [0, \infty],$$

measuring how obstructed T is when x is viewed through r . No interpretation is built in; O is instantiated per domain.

1.2 The two derived quantities (SOIR §3)

Irreducible residue. $I(x; T) := \inf \text{ over } r \in R(x) \text{ of } O(r, x; T)$.

Apparent obstruction. $A(r, x; T) := O(r, x; T) - I(x; T)$, for finite O .

Canonical decomposition (SOIR Thm 4.1). $O = A + I$ with $A \geq 0$: every observed obstruction splits into a representational artifact and an orbit-minimized structural remainder. Trivial to prove, decisive to enforce: it makes the artifact/structure distinction a computed quantity rather than a judgment.

1.3 The three discipline theorems (SOIR §4)

Orbit invariance (Thm 4.2). I depends only on the orbit $R(x)$, never on the representative.

Monotonicity under class restriction (Thm 4.3). $R_2(x) \subseteq R_1(x)$ implies $I \text{ over } R_2 \geq I \text{ over } R_1$. *Consequence, stated by the corpus itself and adopted here as notation:* the residue must always be written indexed to its class — $I_{R(x)}(x; T)$ — because enlarging admissibility can only lower it. Irreducibility is structural relative to declared admissibility, never absolute.

The two licensed routes to positivity (Thm 4.4 + Def 4.5). Pointwise positivity — $O(r) > 0$ for every individual r — does **not** imply $I > 0$, since $O(r_n) \rightarrow 0$ is compatible with it. A positive-residue conclusion requires either **compact attainment** ($R(x)$ compact, O lower semicontinuous, infimum attained at r with $O(r) > 0$) or **quantitative separation** (a single $\delta > 0$ with $O(r) \geq \delta$ for all admissible r). The corpus mandates that every positive claim identify which route is in force. This is the kernel's sharpest tooth, and Section 4.4 below shows it biting in the corpus's favor.

1.4 The classification (SOIR §5)

With a success predicate P over admissible actions $U(x)$, traversability of (x, T) is the existence of (r, u) with P true — logically independent of whether I vanishes. **Four-way classification (Thm 5.2):** exactly one of (1) exactly removable — some r achieves $O = 0$; (2) asymptotically removable — $I = 0$ unattained; (3) irreducible and fatal; (4) irreducible and traversable. Case 4 is the framework's signature regime: a traversal can succeed by organizing itself around what cannot be removed.

1.5 The regime refinements (Appendix BCD)

The obstruction assignment comes in three regimes — strict functor (obstruction-non-increasing morphisms; infima approached monotonically), lax functor (compactness-based attainment; the natural home of gauge phenomena, where invariance of I follows from orbit preservation as a set, **not** from monotonicity — a distinction the appendix correctly refuses to conflate), and function-on-objects (I well-defined but attainment and separation require explicit verification). The four-way classification survives all three regimes.

1.6 The boundary doctrine (BFS)

Every recursively axiomatized system F under resource bound B has a **compression boundary**: statements it cannot resolve (Gödel, Turing, jointly). A **traversal claim** is one derived and asserted within the boundary; a **universality claim** purports to hold beyond it, and requires a witness constitutively unavailable to F . Asserting a universality claim on traversal evidence is itself a semantic obstruction in the SOIR sense — an undeclared representational artifact injected at the meta level. BFS's demonstration that superdeterminism self-undermines is an application; the doctrine itself is the general result.

2. Assessment of the kernel

Tier verdict. Sections 1.1–1.5 are Tier I/II as mathematics: the theorems are elementary (infimum manipulations, Berge attainment) and correctly proved, and the contribution is architectural — a bookkeeping calculus in which the category errors that dominate informal obstruction talk (artifact mistaken for structure; pointwise positivity inflated to positive infimum; class-relative irreducibility asserted absolutely) become *syntactically visible* and therefore uncommittable in silence. Section 1.6 is a correct assembly of classical metamathematics into a usable claim-typing discipline. The calculus is publishable in kernel form once the corrections of Section 3 are applied.

What the kernel is not. It decides nothing by itself. Like the Hopf registry of the traversal, it is chart, not oracle: every substantive conclusion lives in an instantiation — in the choice of X , R , O and in the domain theorems that establish separation or attainment. The corpus knows this (SOIR's prefatory note says it); the consolidation states it as the reading rule.

3. Corrections and demotions

C1 — The Λ -overview formal statement is Tier III and should be demoted to motivation. The root document's displayed claim — artifacts A with $A = \lim f(n) \approx C/L$ and $A \in C \setminus L$, with π and e as artifacts of the curved/linear comparison — is not well-formed as mathematics: C and L are principles, not sets; membership and quotient are undefined on them; and the transcendence of π is a theorem of field theory requiring no C/L ontology. The rigorous descendant of the Λ -intuition is exactly Definition 3.1 plus Theorem 4.3: **irreducibility as an**

indexed infimum. The intuition loses nothing in the demotion; it gains a defensible body. Recommendation: the overview becomes a preface, and the kernel becomes the paper.

C2 — Notation must carry the index. Adopt $I_R(x; T)$ globally. The corpus states the discipline in a remark; a referee will want it in the symbols. Every historical error in this genre is an unindexed I .

C3 — The groupoid draft's lower-bound results inherit the 4.5 obligation. Its universal-lower-bound theorem and pathwise-cost results are sound where the separation hypothesis is explicit; the consolidation flags every use site to name its route (attainment or separation), matching the main paper's standard.

C4 — Terminological collision. "Semantic obstruction" is used in SOIR for the general phenomenon and in BFS for the specific meta-level artifact of undeclared universality. Distinct things. Recommend: *semantic obstruction* (general), *universality artifact* (BFS's specific instance).

4. The flagship instantiation: the Hodge instrument

The Hodge Traversal (Compiled Edition V4–V8) is now exhibited as a full instance of the kernel — and the strongest instance in the combined body of work, because its separation constant is uniform and explicit, where the corpus's own gauge-holonomy example achieves only orbitwise separation.

4.1 The dictionary

Kernel object	Hodge instantiation
State $x \in X$	a pair (A, α) : variety with target Hodge class $\alpha \in \text{Hdg}^k(A)$
Target T	algebraicity: $\mathcal{J}(\alpha) = 0$ in the traversal's residue lens
$R(x)$	the instrument's bridges: $\mathcal{Q}$ -operations applied to (A, α)
Groupoid morphisms	the reframing monoid $(r_{\text{field}}, r_{\text{dim}}, r_{\text{att}}, r_{\text{spec}})$, formally completed to its groupoid of admissible re-descriptions
$O(r, x; T)$	the polarized realization distance, Definition 4.2 below
$I_{\mathcal{Q}}(x; T)$	the instrument-relative irreducible residue
Class enlargement $\mathcal{Q} \subset \mathcal{Q}'$	weld production (carrier change; new filters)

4.2 The obstruction functional, constructed

Definition 4.2 (Realization distance). Fix on $H^{\wedge\{2k\}}(A, \mathbb{Q})$ the metric $\|\cdot\|$ induced by the polarization form and the Hodge structure. For a representation $r \in \mathfrak{L}$ acting on (A, α) , let $\text{Alg}(r) \subseteq H^{\wedge\{2k\}}(A, \mathbb{Q})$ be the set of algebraic cycle classes producible by r (for a G1 reattachment, the cycle classes in its image; for a G2 certificate, $\{0\}$, since certification produces no cycles; for a G3 structure morphism, the cycle classes of its supplied realizations, which by definition of G3 is again $\{0\}$ on the target component). Define

$$O(r, (A, \alpha); \text{alg}) := \inf \{ \|\alpha - c\| : c \in \text{Alg}(r) \}.$$

$O = 0$ exactly when r realizes α as a cycle class; O measures how far the bridge's actual output sits from the target. This makes the traversal's informal "reach of the bridge" a functional in the kernel's sense.

4.3 Theorem (Uniform quantitative separation for the Hodge instrument)

Theorem 4.3. Let (A, α) be any node of the traversal's CM anti-invariant family: α a nonzero Weil-type class in the (-1) -eigenspace V_- of the CM conjugation σ acting on the relevant cohomology. Then for **every** representation $r \in \mathfrak{L}$,

$$O(r, (A, \alpha); \text{alg}) \geq \|\alpha\| > 0,$$

and consequently, by quantitative separation (SOIR Definition 4.5) with the explicit constant $\delta = \|\alpha\|$,

$$I_{\mathfrak{L}}((A, \alpha); \text{alg}) = \|\alpha\| > 0,$$

with the infimum attained at every point of the orbit.

Proof. By the eigenspace obstruction theorem (Compiled Edition, Theorem 17.4), every G1 output lies in V_+ ; G2 and G3 outputs on the target component are $\{0\} \subseteq V_+$ by Definition 4.2. The conjugation σ is an isometric involution with respect to the polarization metric, so $V_+ \perp V_-$. For $c \in V_+$ and $\alpha \in V_-$, $\|\alpha - c\|^2 = \|\alpha\|^2 + \|c\|^2 \geq \|\alpha\|^2$. Hence $O(r) \geq \|\alpha\|$ for all $r \in \mathfrak{L}$, with equality at $c = 0$, which every representation's output set contains or approaches; the bound is attained, and the infimum over the orbit equals $\|\alpha\|$. Closure of the bound under the reframing monoid is Corollary 17.5/17.8 of the Compiled Edition: every generator preserves the hypotheses (a CM pair with isometric conjugation, an anti-invariant target, a conjugation-fixed or non-producing operation). ■

Remark. This is separation in its strongest form — one δ , explicit, uniform over the entire class, attained — strictly stronger than the corpus's holonomy example (SOIR §10.1), where separation holds only orbit-by-orbit with orbit-dependent δ . The Hodge instrument is therefore not merely an application of the calculus; it is its best current exhibit.

4.4 Corollary (Iteration is semantic — the vanishing identity)

Corollary 4.4. For every r in the $\mathfrak{L}$ -orbit of a CM anti-invariant node,

$$A(r, (A, \alpha); \text{alg}) = O(r) - I_{\mathfrak{L}} = \|\alpha\| - \|\alpha\| = \mathbf{0}.$$

The apparent obstruction vanishes identically on the orbit.

Interpretation, and the claim it formalizes. The traversal's repeated finding — "reframing relabels coordinates but does not decrease the residue; iteration is semantic" — is now an *identity in the calculus*: motion along the orbit is motion along a level set of O , and there is no representational artifact ($A \equiv 0$) available to burn off. The phrase "**semantic obstruction is irreducible residue**" thereby acquires its exact technical content, and it is proven for this instrument: within the declared class, the observed obstruction is 100% structural remainder and 0% artifact — not as a verdict earned by exhaustion of patience, but as a computed decomposition under Theorem 4.1. Conversely this supplies the general definition the corpus was reaching for: **a reframing is semantic precisely when it is a morphism along which $dA = 0$ and $dI = 0$; it is structural precisely when it changes the admissible class**. Semantic/structural is thus not a judgment call; it is the orbit/class distinction, functionalized.

4.5 Theorem (Historical strict monotonicity — the Markman witness)

Theorem 4.5. Let $\mathcal{Q}$ be the instrument class and $\mathcal{Q}' \supset \mathcal{Q}$ its enlargement by hyperholomorphic/semiregular sheaf representations on hyper-Kähler carriers (the Markman-era technology). For fourfold Weil nodes (A, α) :

$$I_{\mathcal{Q}}((A, \alpha); \text{alg}) = \|\alpha\| > 0 \text{ while } I_{\mathcal{Q}'}((A, \alpha); \text{alg}) = 0,$$

with the zero **exactly attained** (case 1 of the four-way classification): the published constructions supply witness representations realizing α as an algebraic cycle class, for all imaginary quadratic fields and all discriminants.

Proof. The left value is Theorem 4.3. The right value is the literature weld recorded in the Compiled Edition §22 (Markman; Floccari; Fu–Floccari): the sheaf-theoretic representations are not conjugation-fixed, their outputs land in V_- , and the constructions terminate in explicit cycle classes, so $\text{Alg}(r_{\text{Markman}}) \ni \alpha$ and $O(r_{\text{Markman}}) = 0$. ■

Interpretation. SOIR's Theorem 4.3 (monotonicity under class restriction) now has a strict-inequality witness of Millennium-problem grade, dated and citable: the residue dropped from a proven positive constant to an attained zero when, and only when, the admissible class was enlarged. The corpus's discipline remark — *the residue is structural relative to declared admissibility, not absolute* — is no longer a caution; it is a documented historical event. No better validation of the calculus exists or is likely to.

4.6 Theorem (The bar on the absolute claim, derived from the corpus's own kernel)

Theorem 4.6. Let $\mathcal{Q}''$ denote the class of all representations available in current mathematics, and let (A, α) be a surviving sixfold node (discriminant $\neq -1$; the McMullen family as sharpest member). Then:

1. $O(r, (A, \alpha); \text{alg}) > 0$ for every $r \in \mathcal{Q}''$ — this is the corpus exhaustion of the Compiled Edition (Main Theorem B; the family-specific inaccessibility theorem), i.e. **pointwise positivity over the entire current class**;
2. nevertheless, **no positive-residue conclusion $I_{\mathcal{Q}''} > 0$ is derivable for any class $\mathcal{Q}'''$ containing all future mathematics**, because both licensed routes of SOIR §4.4 fail for such a class: it is not compact — it is not even a definable totality (quantifier-exhaustion theorem, cell 4), so attainment arguments cannot be posed on it — and quantitative separation over it would require a necessary condition of algebraicity that a target violates, which the filter inventory proves does not exist and whose construction would refute the motivated-implies-algebraic architecture;

3. therefore the assertion "the Hodge residue is absolutely irreducible" is, in BFS's typology, a **universality claim asserted on traversal evidence** — the exact malformation BFS names — and the corpus's own kernel is what proves it malformed.

Proof. (1) is cited. (2): non-definability of the class is the Compiled Edition's Theorem 30(E) cell 4; failure of separation is Main Theorem B plus component C's pricing of the missing filter. (3) is Definition 3.2 of BFS applied to the statement in question, whose scope quantifies past the compression boundary of every fixed system by construction. ■

Interpretation — the loop closed. Read jointly, SOIR and BFS *derive* the discipline that the entire Hodge Compiled Edition enforced turn by turn: the demand "prove the residue absolutely irreducible, or you have not exhausted the frontier" is not merely unmet — it is proven, inside the corpus's own calculus, to be a demand for a universality claim that no bounded system can witness, whose licensed substitutes (attainment, separation) are individually and provably unavailable, and whose historical track record (Theorem 4.5) shows the unindexed version to have been *false* the one time it was testable. The framework did not need to borrow the compiled edition's conclusions; it independently forces them. That is what it means for the repository and the traversal to be one body of work.

4.7 The classification, populated

The four-way classification (SOIR Thm 5.2), instantiated over the Hodge node family, with every cell inhabited or provably problematic to inhabit:

Case	Definition	Hodge inhabitants
1 — exactly removable	some r achieves $O = 0$	$\mathbb{CP}^n$; K3 degree two (Lefschetz (1,1)); Weil fourfolds, all fields and discriminants (Markman era); Kuga–Satake on the generalized-Kummer locus and cited K3 families
2 — asymptotically removable	$I = 0$ unattained	no known Hodge instance ; plausibly empty here, since the natural O of Definition 4.2 on the natural compactifications is lower semicontinuous, forcing attainment — recorded as an open structural question of the calculus itself
3 — irreducible and fatal	$I_R > 0$, not traversable	the CM anti-invariant family relative to $\mathcal{Q}$ with T read as "decide via $\mathcal{Q}$ ": $I_{\mathcal{Q}} = \ d\ $ and no $\mathcal{Q}$ -action succeeds (Compiled Edition Thm 20.1)
4 — irreducible and traversable	$I_R > 0$, traversable	the same family relative to $\mathcal{Q}$ with T read as the traversal's actual practice: the program succeeds — closes nodes, relocates frontiers, prices doors — while organizing itself around the unremoved residue; and historically, case-3 nodes migrate to case 1 under class enlargement, which is the observed dynamics of the field

The migration pattern $3 \rightarrow 1$ under $\mathcal{L} \subset \mathcal{L}'$ (never $3 \rightarrow 2$, never $1 \rightarrow 3$; monotonicity forbids regressions) is the calculus's prediction for the frontier's future, and it matches every recorded event of the modern era.

5. Standing

What this consolidation asserts as proven: the kernel of Sections 1.2–1.4 as restated; Theorems 4.3, 4.5, 4.6 and Corollary 4.4 as new results of the joint body of work, each reducible to the cited components of the Compiled Edition plus elementary arguments given here in full. What it recommends as editorial action on the repository: the demotion C1, the notation C2, the flag C3, the terminology C4 — after which the kernel paper (SOIR + BCD, with Section 4 of this document as its flagship application) is, in the consolidator's judgment, the publishable spine of the corpus. What it does not claim: any change to the status of the Hodge conjecture, whose open cell, pending finite computation, and one-directional traffic are exactly as recorded in the Compiled Edition — now with the additional statement, computed rather than argued, that within every class where the question can be posed, the obstruction observed is residue entire, artifact nil.